

$$= \frac{2n^2 + 2n + 1}{6} - \frac{n^2 + 2n + 1}{4}$$

$$= \frac{4n^2 + 6n + 2 - 3n^2 - 6n - 3}{12}$$

$$= \frac{n^2 - 1}{12}$$

Discrete distribution :

① Binomial distribution : Let X be a discrete random variable taking non-negative values, then X is said to be follow binomial distribution if it has pmf :

$$P[X=r] = {}^nC_r (p)^r (q)^{n-r}, \quad p+q=1$$

n = no. of trials / experiment

p = prob of success

q = prob of failure

r = no. of success

⊗ Mean = $E[X] = np$

⊗ Variance = npq

Q. A die is rolled 15 times, what is the prob of getting 2 or 3, 10 times ?

Solⁿ: $n=15$ $P(2 \text{ or } 3) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$

Let getting 2 or 3 be a success,

$$p = \frac{1}{3}, \quad q = \frac{2}{3}, \quad n = 15, \quad r = 10$$

$$P[X=10] = {}^{15}C_{10} \left(\frac{1}{3}\right)^{10} \left(\frac{2}{3}\right)^5 = {}^{15}C_{10} \frac{2^5}{3^{15}}$$

Q. A & B play a game in which their chance of winning are in ratio 3:2 find A's chance of winning atleast 3 games out of 5 games played.

Solⁿ: $P(\text{winner is A}) = 3$ $p = \frac{3}{5}$, $q = \frac{2}{5}$, $n = 5$, $r = 3$
 $P(\text{winner is B}) = 2$

$$P[X=3] = {}^5C_3 \left(\frac{3}{5}\right)^3 \left(\frac{2}{5}\right)^2 = {}^5C_3 \frac{27 \times 4}{(5)^5}$$

$$= \frac{5!}{3! \times 2!} \times \frac{27 \times 4}{(5)^5} = 0.3456$$

$$P[X=4] = {}^5C_4 \left(\frac{3}{5}\right)^4 \left(\frac{2}{5}\right)^1 = \frac{5 \times 3 \times 3 \times 3 \times 3 \times 2}{3125} = 0.2592$$

$$P[X=5] = {}^5C_5 \left(\frac{3}{5}\right)^5 \left(\frac{2}{5}\right)^0 = \frac{3 \times 3 \times 3 \times 3 \times 3}{3125} = 0.07776$$

$$\Rightarrow 0.3456 + 0.2592 + 0.07776$$

$$\Rightarrow 0.6825$$

Q. In a binomial distribution consisting of 5 independent trials, prob of '1' & '2' success are 0.4096 & 0.2048 respectively. find parameter, p of distribution

Solⁿ: $n = 5$, $p = ?$, $r_1 = 1$, $r_2 = 2$

for x_1

$$P[X=1] = {}^5C_1 (p)^1 (q)^4 = {}^5C_1 (p)^1 (1-p)^4 = 0.4096 \quad \text{--- ①}$$

$$\text{for } x_2: P[X=2] = {}^5C_2 (p)^2 (q)^3 = {}^5C_2 (p)^2 (1-p)^3 = 0.2048 \quad \text{--- ②}$$

$$\frac{\text{①}}{\text{②}} \quad \frac{5 \cdot p \cdot (1-p)^4}{10 \cdot p^2 \cdot (1-p)^3} = 2 \Rightarrow 1-p = 4p$$

$$p = \frac{1}{5}$$

~~$P[X > 10] = 1 - \dots$~~

② Poisson Distribution: A discrete random variable X follows poisson's distribution if it has the pmf

$$P[X=x] = \frac{e^{-\lambda} \cdot \lambda^x}{x!}, \quad \lambda = \text{mean} = np$$

large less

Q. A manufacturer of bulb knows that 5% of his product defective. If he sells bulbs in boxes of 100 & guarantees that not more than 10 bulbs will be defected. What is the approximate probability that a box will fail to meet the guaranteed quality.

Solⁿ:

$$p = 5\% = 0.05, \quad \lambda = np = 5$$
$$n = 100$$

$$P[X > 10] = 1 - P[X \leq 10]$$
$$= 1 - \sum_{x=0}^{10} \frac{e^{-5} (5)^x}{x!}$$

Q. A car hire firm has 2 cars which it hires out day by day. The no. of demands for a car on each day is distributed as poisson's distribution with mean 1.5. Calculate proportion of days on which

- (a) neither car is used
- (b) some demand is refused

Soln:

② $X = \text{no. of car used}$ $\lambda = 1.5$
$$P[X=0] = \frac{e^{-1.5} (1.5)^0}{0!} = \frac{e^{-1.5} (1.5)^0}{0!} = 0.223$$

⑤ $P[X > 2] = 1 - P[X \leq 2]$
$$= 1 - \sum_{x=0}^2 \frac{e^{-1.5} (1.5)^x}{x!} = 0.1912$$

③ Geometric Distribution: A discrete random variable taking non-negative values follows geometric distribution if it has pmf.

$$P[X=r] = q^{r-1} p$$

where $P[X=r]$ is regarded as getting first success on r^{th} trial.

Note: $E[X] = \text{mean} = 1/p$

$$\text{var}[X] = \frac{q}{p^2}$$

Q. If the prob that the target is destroyed on any one shot is 0.5. What is the prob that it will be destroyed on 6th attempt.

Soln: $P[X=r] = q^{r-1} p = (0.5)^5 (0.5)$
$$= \left(\frac{1}{2}\right)^6 = \frac{1}{64}$$

Q. If prob that an applicant for a driver's license will pass road test on any given trial is 0.8. What is the prob that he will finally pass test

- i) on the 4th trial
- ii) fewer than 4 trials

Solⁿ:

$$P[X=4] = (0.2)^3(0.8) = 0.0064$$

$$\begin{aligned} P[X \leq 4] &= P[X=1] + P[X=2] + P[X=3] \\ &= (0.2)^0(0.8) + (0.2)^1(0.8) + (0.2)^2(0.8) \\ &= 0.992 \end{aligned}$$

Q. If the prob of success on each trial is 0.25 after how many trials can be expect first success.

Solⁿ: $p = 0.25, q = 0.75$

$$E[X] = \frac{1}{0.25} = 4$$

Q. The mean & variance of binomial distribution are 4 & $\frac{4}{3}$ respectively. find prob of $X > 1$

Solⁿ:
$$\frac{np = 4}{npq = \frac{4}{3}} \Rightarrow \frac{1}{q} = 3 \text{ \& } p = \frac{2}{3}$$

$$n \cdot \frac{2}{3} = 4 \Rightarrow \boxed{n=6}$$

$$\begin{aligned} P[X > 1] &= 1 - P[X \leq 1] \\ &= 1 - P[X=0] \\ &= 1 - {}^6C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^6 \\ &= 1 - \frac{1}{729} = \frac{728}{729} \end{aligned}$$

Continuous Distribution:

① Normal distribution: A CRV X follows normal distribution with parameter μ & σ if it has pdf

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad -\infty \leq x \leq \infty$$

μ & σ are mean & SD of this distribution respd after standardization:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z)^2} \quad \text{standard normal distribution}$$

$\mu=0, \sigma=1$

Properties:

① Graph of pdf of normal distribution is bell shaped curved.

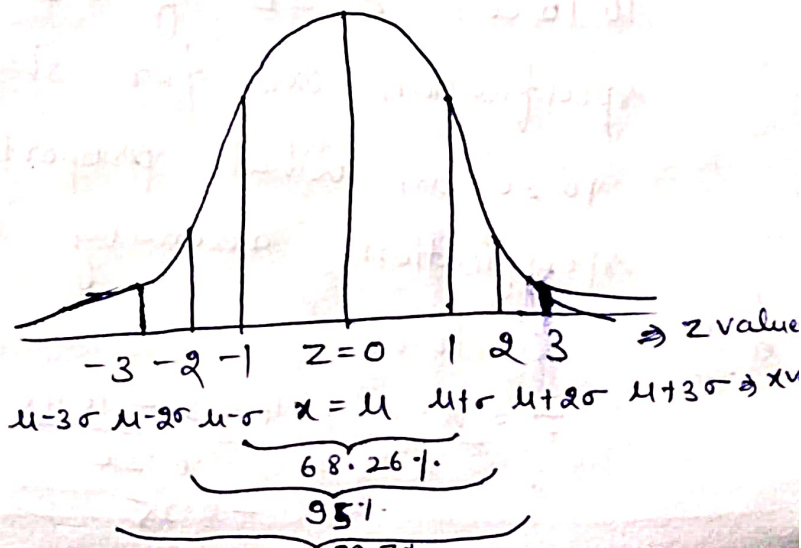
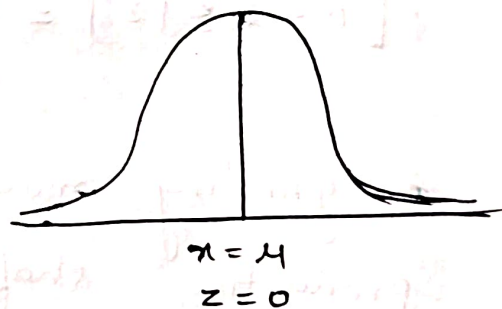
② Graph is symmetric about the line $x=\mu$ (or $z=0$)

③ Mean, median & mode of normal distribution coincide

④ Empirical formula

$$P[-1 \leq Z < 1] = 0.6826$$

$$P[0 \leq Z \leq 1] = 0.3413$$



Q. A CRV X is normally distributed with $\mu = 12$ & SD 4 respectively. find following
① $P[X > 20]$ ② $P[X \leq 20]$ ③ $P[0 \leq X \leq 12]$

Soln.

① $\mu = 12, \sigma = 4$

$$X = 20 \Rightarrow Z = \frac{20 - 12}{4} = 2$$

$$P[X > 20] = P[Z > 2] = 0.5 - P[0 \leq Z \leq 2] \\ = 0.5 - \frac{0.95}{2} = 0.025$$

② $X = 20, Z = \frac{20 - 12}{4} = 2$

$$P[X \leq 20] = P[Z \leq 2] = 1 - P[Z > 2] \\ = 1 - 0.025 = 0.975$$

③ $X = 0$

$$X = 12$$

$$Z = \frac{0 - 12}{4} = -3$$

$$Z = \frac{12 - 12}{4} = 0$$

$$P[0 \leq X \leq 12] = P[-3 \leq Z \leq 0] = \frac{0.997}{2}$$

Q. A grinding machine is so set that its production shafts have an avg diameter 10.10 cm & SD of 0.20 cm. The product specification call for shaft diameter b/w 10.05 & 10.20 cm. What proportion of output meets the specifications assuming normal distribution.

Soln.

$$X = 10.05$$

$$Z = \frac{X - \mu}{\sigma} = \frac{10.05 - 10.10}{0.20}$$

$$= -0.25$$

$$X = 10.20$$

$$Z = \frac{10.20 - 10.10}{0.20}$$

$$= 0.5$$

$$P[-0.25 \leq Z \leq 0.5]$$

$$\downarrow \qquad \qquad \downarrow$$

$$0.40129 \qquad 0.69146$$

$$0.69146 - 0.40129 = \underline{0.29017}$$

Q. X is a normal distribution funcⁿ with mean 30 & SD 4 find prob of i) $X < 40$ ii) $P[30 < X < 35]$ iii) $P[X > 21]$

Solⁿ:

① $X < 40$

$$Z = \frac{40 - 30}{4} = 2.5$$

$$Z < 2.5 = 0.99379$$

② $X = 30$

$$Z = 0$$

$X = 35$

$$Z = \frac{35 - 30}{4} = 1.25$$

$$P[0 \leq Z \leq 1.25]$$

$$\downarrow \qquad \qquad \downarrow$$

$$0.500 \qquad 0.89435$$

$$= 0.39435$$

③ $X = 21$

$$Z = \frac{21 - 30}{4} = -2.25$$

$$Z > -2.25 = 0.1222$$

Q. X is normally distribution with mean 12 & SD 4 find value of x_0 s.t $P[X > x_0] = 0.24$

Solⁿ:

$$P[X > x_0] = 0.24$$

$$1 - P[Z \leq z_0] = 0.24$$

$$P[Z \leq z_0] = 0.76$$

$$z_0 = 0.71$$

$$Z = \frac{X - \mu}{\sigma}$$

$$0.71 = \frac{X - 12}{4}$$

$$X = 14.84$$

Q. A normal dist 31% of items are under 45 & 8% are over 64 find mean & SD.

Solⁿ:



$$P[Z \leq z_0] = 0.31 \quad \& \quad P[Z \leq z_1] = 0.92$$

$$z_0 = -0.49$$

$$z_1 = +1.41$$

$$\frac{45 - \mu}{\sigma} = -0.49$$

$$45 - \mu = -0.49\sigma$$

①

$$\frac{64 - \mu}{\sigma} = +1.41$$

$$64 - \mu = +1.41\sigma$$

②

Solving ① & ②

$$45 - \mu = -0.49\sigma$$

$$64 - \mu = 1.41\sigma$$

$$\hline -19 = -1.9\sigma$$

$$\frac{19}{1.9} = \sigma = 10$$

$$64 - \mu = 1.41(10)$$

$$64 - 14.1 = \mu$$

$$\boxed{\mu = 50}$$

Central Limit Theorem: Let $x_1, x_2, \dots, x_n$ be a random sample taken from a population following any distribution. Then for a sufficiently large n , the distribution of sample mean follows normal distribution neither mean equals to population mean and SD equals to $\frac{1}{\sqrt{n}}$ times SD of population i.e., $\mu_{\bar{x}} = \mu$ & $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$

Q. Given a population with mean of 110 & SD is 20. What is prob that the mean of random sample of 25 will be greater than 120.

Solⁿ:

$$\mu_{\bar{x}} = \mu = 110$$

$$\sigma_{\bar{x}} = \frac{1}{\sqrt{25}} (20) = 4$$

$$P[\bar{X} > 120] = 1 - P[Z < 2.5] = 1 - 0.993 = 0.0062 \quad \text{Ans}$$

$$Z = \frac{X - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{120 - 110}{4} = 2.5$$

Q. An inspector knows from past experience that the time it takes to grade one student paper is heavily skewed (non-normal), has an avg of 6 min & SD of 7 min. This term she has a class of 49 students, she decided to budget an avg of 8 min per student paper for grading. What is the likelihood she has not budgeted enough time? In other words, what is the prob that the sample avg will be greater than 8 min.

Soln:

$$\mu_{\bar{x}} = 6$$

$$\sigma_{\bar{x}} = \frac{1}{\sqrt{49}} (7) = 1$$

$$Z = \frac{x - \mu}{\sigma} = \frac{8 - 6}{1} = 2$$

$$\Rightarrow 1 - P[Z < 2] = 1 - 0.97725 = \underline{0.02275}$$

Q. Exponential Distribution: Let X be continuous random variable. Then X follows exponential distribution with parameter θ if it has the pdf.

$$f(x) = \begin{cases} \theta e^{-\theta x}, & x \geq 0 \\ 0, & \text{otherwise} \end{cases} \quad \text{where } \theta > 0$$

Note: Mean = $\frac{1}{\theta}$

$$\text{Variance} = \frac{1}{\theta^2}$$

Q. A mobile conversation follows exp distri with pdf $f(x) = \frac{1}{3} e^{-\frac{1}{3}x}$. What is the prob that the conversation takes more than 5 min.

Soln:

$$\theta = \frac{1}{3}$$

$$\text{Mean} = \frac{1}{\theta} = 3$$

$$\text{Variance} = 9$$

$$\text{SD} = 3$$

$$Z = \frac{x - \mu}{\sigma}$$

$$= \frac{5 - 3}{3} = 0.66$$

$$P[X \geq 5] = 1 - P[Z < 0.66]$$
$$= 1 - 0.74537$$
$$= 0.2546$$

$$= \int_5^{\infty} \frac{1}{3} e^{-\frac{1}{3}x} dx$$

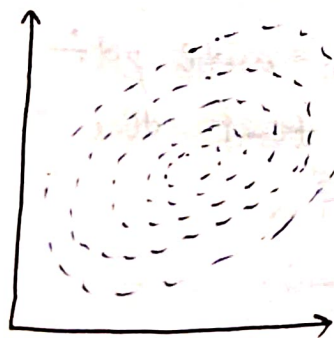
Co-relation : If a change in one variable leads to the change in other variables, then the variables are said to be correlated.

⇒ If an increase in the value of one variable corresponds to an increase (or decrease) in other variable, then correlation is said to be positive.

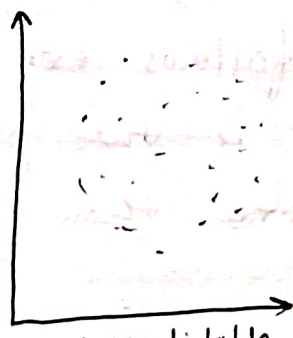
⇒ If an increase (or decrease) in the value of one variable corresponds to decrease (or increase) in other variable, the correlation is negative.

⇒ If there is no relationship b/w 2 variables, they are said to be uncorrelated (or independent).

If 2 variables vary in such a way that their ratio is always constant then correlation is perfect



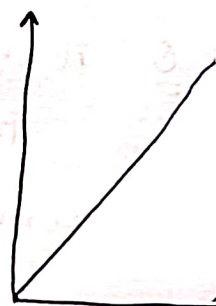
+ve



unpredictable



-ve



perfect

Karl - Pearson's coefficient of correlation :

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

<u>Q</u>	X	Y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$	$(y - \bar{y})^2$
	2	7	-2	-4	8	4	16
	3	9	-1	-2	2	1	4
	4	10	0	-1	0	0	1
	5	14	1	3	3	1	9
	6	15	2	4	8	4	16
					<u>21</u>	<u>10</u>	<u>46</u>

$$\bar{x} = 4$$

$$\bar{y} = 11$$

$$r = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sqrt{\sum (x - \bar{x})^2 \sum (y - \bar{y})^2}}$$

$$= \frac{21}{\sqrt{10 \times 46}} = \frac{21}{21.44} = 0.9791$$

Alternate Formulation :

$$1) \quad r = \frac{\sum xy}{\sqrt{\sum x^2 \sum y^2}}$$

where x & y are deviations about the mean

$$2) \quad r = \frac{\sum xy}{n \sigma_x \sigma_y} \quad \text{or} \quad \frac{\sum (x - \bar{x})(y - \bar{y})}{n \sigma_x \sigma_y}$$

$n = \text{no. of observation}$

$$3) \quad r = \frac{\sum xy - \frac{\sum x \sum y}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n} \right) \left(\sum y^2 - \frac{(\sum y)^2}{n} \right)}}$$

Q. Calculate correlation coeff b/w x & y using following data $n=12$, $\Sigma x=96$, $\Sigma (x-8)^2=150$, $\Sigma (y-10)^2=200$
 $\Sigma (x-8)(y-10)=50$

Soln:

$$r = \frac{\Sigma (x-\bar{x})(y-\bar{y})}{\sqrt{\Sigma (x-\bar{x})^2 \Sigma (y-\bar{y})^2}} = \frac{50}{\sqrt{150 \times 200}} = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6}$$

Q. A person while calculating r b/w two variables x & y . obtain following result $n=30$, $\Sigma x=120$,
 $\Sigma x^2=600$, $\Sigma y=90$, $\Sigma y^2=250$, $\Sigma xy=356$.

At the time of checking the calculation that 2 pairs of obs $(8,10)$ & $(12,7)$ were wrongly entered instead $(8,12)$ & $(10,8)$. Determine correct value of coeff of correlation.

Soln:

$$r = \frac{\Sigma xy - \frac{\Sigma x \Sigma y}{n}}{\sqrt{\left(\Sigma x^2 - \frac{(\Sigma x)^2}{n}\right) \left(\Sigma y^2 - \frac{(\Sigma y)^2}{n}\right)}} = \frac{356 - \frac{120 \times 90}{30}}{\sqrt{\left(600 - \frac{(120)^2}{30}\right) \left(250 - \frac{(90)^2}{30}\right)}} = 0.14$$

Q. find correlation coefficient $n=30$, $\Sigma x^2=190$, $\Sigma y^2=193$
 $\Sigma xy=117$ where x, y is deviation from mean.

Soln:

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \Sigma y^2}} = \frac{117}{\sqrt{190 \times 193}} = 0.610 \text{ (+ve strong correlation)}$$

Types of correlation:

- ① If $r > 0.5 \Rightarrow$ strong correlation ($0.5 \rightarrow 1$)
- ② If $r < 0.5 \Rightarrow$ weak correlation ($0 \rightarrow 0.5$)
- ③ If $r = 0.5 \Rightarrow$ moderate correlation
- ④ If $r = 1$ or $-1 \Rightarrow$ perfect correlation

Regression Analysis :

1. Simple linear regression :

It is a statistical tool or method used to model relationship b/w two variables : dependent & independent.

- for predicting or finding a mathematical eqn by using ordinary least square estimation method in which we minimize square of deviation.

i.e.,
$$y = \beta_0 + \beta_1 x + e_i$$

y = dependent, β_1 = slope

x = independent, e_i = error or residual

β_0 = intercept

$$e_i = y_{\text{obs}} - y_{\text{pred}}$$

Assumption: e_i follows normal distribution

2. Multiple linear regression :

If there are more than one independent variable

i.e.,

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + e_i$$

Coefficient of determination (R^2):

It is square of correlation coeff (r)

$$R^2 = (r)^2 \text{ i.e., } 0 \leq R^2 \leq 1$$

$\Rightarrow$ If $R^2 \geq 90\%$ i.e., 90% variation in y (dependent) is explained by x (independent) to the model

$\Rightarrow$ If $R^2 \geq 60\%$ i.e., the model is best fit

Regression lines / eqn:

① $\bar{x}$ & $\bar{y}$ always lie on regression line.

② Slope of regression lines are:

$$b_{yx} = r \frac{\sigma_y}{\sigma_x} = y \text{ on } x$$

$$b_{xy} = r \frac{\sigma_x}{\sigma_y} = x \text{ on } y$$

③ Eqn of regression line:

* y on x

$$y - \bar{y} = b_{yx}(x - \bar{x}) = r \frac{\sigma_y}{\sigma_x}(x - \bar{x})$$

* x on y

$$x - \bar{x} = b_{xy}(y - \bar{y}) = r \frac{\sigma_x}{\sigma_y}(y - \bar{y})$$

$$④ \quad b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = r \frac{\sigma_y}{\sigma_x}$$

$$b_{xy} = \frac{n \sum xy - \sum x \sum y}{n \sum y^2 - (\sum y)^2} = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (y - \bar{y})^2} = r \frac{\sigma_x}{\sigma_y}$$

⑤ $r = \sqrt{b_{xy} \cdot b_{yx}}$

sign of r will be same as sign of b_{yx} & b_{xy}

Q.

x	y	xy
1	9	9
2	8	16
3	10	30
4	12	48
5	11	55
6	13	78
7	14	98
8	16	128
9	15	135
		<u>597</u>

soln:

find regression lines.

$$\bar{x} = 5$$

$$\bar{y} = 12$$

$$b_{xy} = 0.95$$

$$y - \bar{y} = b_{yx}(x - \bar{x})$$

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$= \frac{9(597) - 4860}{9(285) - (2025)}$$

$$= 0.95$$

$$y - 12 = 0.95(x - 5)$$

$$\boxed{y = 0.95x + 7.25}$$

$$x - \bar{x} = b_{xy}(y - \bar{y})$$

$$x - 5 = 0.95(y - 12)$$

$$\boxed{x = 0.95y - 6.4}$$

★★

Q.

given the following regression lines y on x & x on y as $3x + 4y = 7$ & $4x + y = 5$ respectively. find correlation coeff. b/w x & y & their means.

soln:

$$y = \frac{7}{4} - \frac{3}{4}x \quad \& \quad x = \frac{5}{4} - \frac{1}{4}y$$

$$b_{yx} = -\frac{3}{4} \quad \& \quad b_{xy} = -\frac{1}{4}$$

$$r^2 = \left(-\frac{3}{4}\right)\left(-\frac{1}{4}\right) \Rightarrow -\frac{3}{4}$$

$$3\bar{x} + 4\bar{y} = 7$$

$$16\bar{x} + 4\bar{y} = 20$$

$$-13\bar{x} = -13$$

$$\boxed{\bar{x} = 1}$$

$$\boxed{\bar{y} = 1}$$

Q. Given the following values, find expected value of x when $y = 12$, $r = 0.42$

	x	y
Avg	25	22
SD	4	5

Soln:

$$x - \bar{x} = b_{ny}(y - \bar{y})$$

$$x - 25 = b_{ny}(y - 22)$$

$$\frac{x - 25}{-10} = b_{ny}$$

$$b_{ny} = r \frac{\sigma_x}{\sigma_y} = 0.42 \times \frac{4}{5} = 0.336$$

$$\frac{x - 25}{-10} = 0.336 \Rightarrow \boxed{x = 21.64}$$

Q. Estimate the regression of y on x given that $\sum x = 15$, $\sum y = 15$, $\sum x^2 = 49$, $\sum y^2 = 49$, $\sum xy = 44$, $n = 5$

Soln:

$$b_{yx} = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2} = \frac{5(44) - (15)(15)}{5(49) - (15)^2} = -0.25$$

$$\bar{x} = \frac{\sum x}{n} = \frac{15}{5} = 3$$

$$(y - 3) = -\frac{1}{4}(x - 3)$$

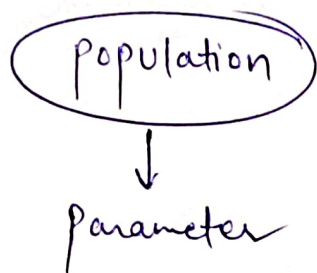
$$\bar{y} = \frac{\sum y}{n} = \frac{15}{5} = 3$$

$$\boxed{4y + x - 15 = 0}$$

Testing of hypothesis :

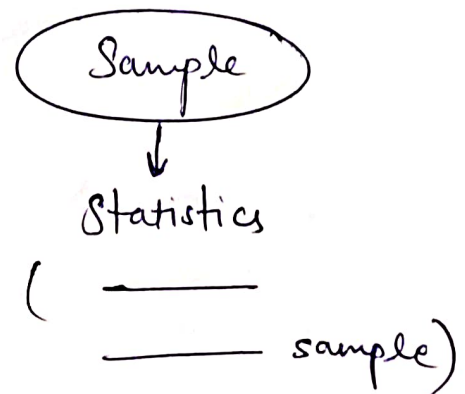
1. Population : A well defined collection of individuals/ object or set of all experimental data.
2. Sample : A small section selected from the population i.e., a finite subset of the population is called sample.
⇒ The no. of individuals in a sample is sample size
⇒ The process of drawing sample is sampling

3. Parameter & Statistics



(measures used to describe the population)

- ① population mean - μ
- ② population SD - σ
- ③ population size - N
- ④ population proportion - p
- ⑤ population variance - σ^2



- ① sample mean - $\bar{x}$
- ② sample SD - s
- ③ sample size - n
- ④ sample proportion - p
- ⑤ sample variance - s^2

Note : The object of sampling studies is to obtain best possible value of the parameter under specific condition. The logic of sampling theory is the logic of induction in which we pass from a particular (sample) to general (population) - Such generalization from sample to pop is called statistical inference

4. classification of sample :

(a) large sample : If the size of sample
 $n \geq 30$

(b) Small sample : If the size of sample
 $n < 30$

Hypothesis : It is required to make decisions about population on the basis of sample information.

Test of hypothesis : Deciding a hypothesis is to be accepted or rejected.

Null hypothesis : The hypothesis formulated for the shape of rejecting it, under the assumption that it is true.

It is usually denoted by H_0 .

Alternate hypothesis : Any hypothesis which is complementary to the null hypothesis.

It is denoted by H_1 .

Testing of significance : The procedure / test which enables us to decide whether the null hypothesis is accepted or rejected on the basis of sample information.

Level of significance : It is the probability level below which we reject the hypothesis.

Errors in Sampling : The main objective in the sampling theory is to draw conclusions or valid inferences about the population parameter on the basis of results of sample. In this process we are liable to commit the following 2 types of error.

- ① Type-I error : Reject H_0 while it is true
- ② Type-II error : Accept H_0 while it is false

Decision Table :

	Accept H_0	Reject H_0
H_0 is true	Correct decision	Type I error
H_0 is false	Type II error	Correct decision

Procedure for testing of hypothesis :

1. Setup the null hypothesis H_0 which is suitable to problem under steady condⁿ.
2. Setup the alternate hypo H_1 complementary to the null hypothesis H_0 .
3. Choose the appropriate level of significance in advance.
4. Compute the value of test statistics.
5. Comparison & conclusion : Compare the calculated value obtained from previous step with some given value & then draw conclusion.

Type of test :

χ^2 -test (chisquare test) :

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

$O_i \rightarrow$ observed value

$E_i \rightarrow$ expected value.

Q. from the following table regarding color of eyes of father & son, test if color of son's eye is associated with that of father. The value of χ^2 is 5% level of significance is 3.841

		eye color of son		
		light	not light	
eye color of father	light	471	51	522
	not light	148	230	378
		619	281	900

solⁿ:

$$E_1 = \frac{522 \times 619}{900} = 359.02$$

$$E_2 = \frac{522 \times 281}{900} = 162.98$$

$$E_3 = \frac{378 \times 619}{900} = 259.98$$

$$E_4 = \frac{378 \times 281}{900} = 118.02$$

$$\chi^2 = \frac{(471 - 359.02)^2}{359.02} + \frac{(51 - 162.98)^2}{162.98} + \frac{(148 - 259.98)^2}{259.98} + \frac{(230 - 118.02)^2}{118.02}$$

$$\chi^2 = 266.336$$

If $\chi^2_{cal} > \chi^2_{tab} \Rightarrow H_0$ is rejected

$$266.3 > 3.84 \Rightarrow H_0 \text{ is rejected}$$

<u>Q.</u> day shifts	good parts	bad parts	
evening shift	940	50	- 950
morning shift	950	45	- 995
day shift	960	40	- 1000
	<u>2850</u>	<u>135</u>	<u>2945</u>

soln: $E_1 = \frac{2850 \times 1000}{2945} = 967.74$

$$E_2 = \frac{135 \times 1000}{2945} = 45.84$$

$$E_3 = \frac{2850 \times 950}{2945} = 919.35$$

$$E_4 = \frac{135 \times 950}{2945} = 43.5$$

$$E_5 = \frac{2850 \times 995}{2945} = 962.9$$

$$E_6 = \frac{995 \times 135}{2945} = 45.6$$

$$\chi^2 = \frac{(960 - 967.74)^2}{967.74} + \dots$$

$$\chi^2 = 1.27$$

$$\chi_{cal}^2 < \chi_{tab}^2$$

$\Rightarrow H_0$ is failed to reject.

$\Rightarrow$ production of body parts is independent of shift

Test Criteria

Large sample

($n \geq 30$)

Z-test

Small sample

($n < 30$)

t-test

① One-population mean

② Two-population mean

a) Independent t-test

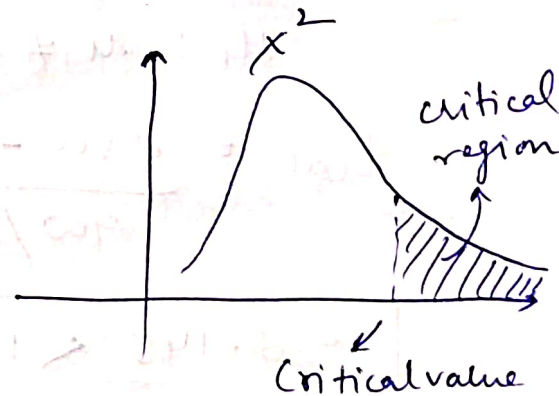
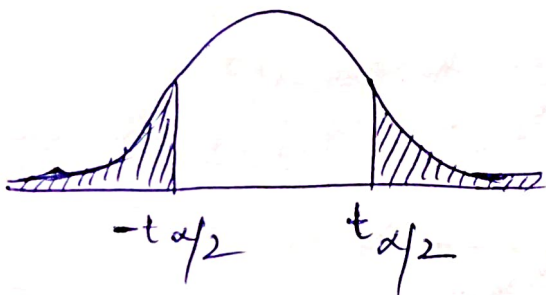
b) Dependent t-test

① One-population mean :

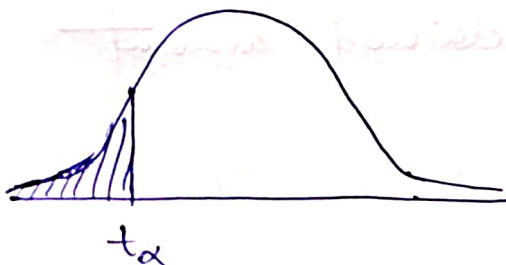
$H_0 : \mu = \mu_0$, $H_0 : \mu \geq \mu_0$, $H_0 : \mu \leq \mu_0$

$H_1 : \mu \neq \mu_0$ $H_1 : \mu < \mu_0$ $H_1 : \mu > \mu_0$

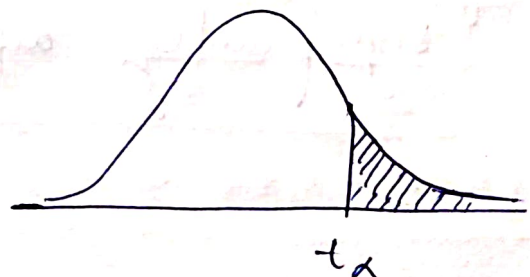
① Two tailed :



② Left tailed :



③ Right tailed :



$$t_{cal} = \frac{\bar{x} - \mu}{s/\sqrt{n}}, \quad s = \sqrt{\frac{1}{n-1} \sum (x_i - \bar{x})^2}$$

Q. A researcher claims that avg daily sodium intake of adults in a city is 2300 mg. To test this claim, a researcher collects a random sample of 15 adults & finds that their avg sodium intake is 2500 mg with a SD of 400 mg. At 5% level of significance test whether the avg sodium intake is significantly diff from claimed amount with critical value $= \pm 2.145$

Soln: $\bar{x} = 2500, n = 15$
 $\mu = 2300, s = 400$

$$H_0: \mu = 2300$$

$$H_1: \mu \neq 2300$$

$$t_{cal} = \frac{2500 - 2300}{400/\sqrt{15}} = 1.93$$

$$-2.145 < 1.93 < 2.145$$

$\Rightarrow H_0$ is accepted

$\Rightarrow$ avg sodium intake in this city is not significantly diff from claimed amount

② Two population mean

Independent t-test

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$H_0: \mu_I \leq \mu_{II}, H_0: \mu_I \geq \mu_{II}$$

$$H_1: \mu_I > \mu_{II}, H_1: \mu_I < \mu_{II}$$

Dependent t-test

$$t = \frac{\bar{d} - d_0}{SD/\sqrt{n}}$$

$\bar{d}$ = mean of diff

d_0 = measurement of changes

n = sample size

$$H_0: \mu_d \leq 0, H_0: \mu_d \geq 0$$

$$H_1: \mu_d > 0, H_1: \mu_d < 0$$

Q. In a certain experiment to compare two types of animal food: A & B the following results of increase in weights were observed in animals

A: 49, 53, 51, 52, 47, 50, 52, 53

B: 52, 55, 52, 53, 50, 54, 54, 53

Assuming that two sample of animal are dependent can be conclude that food B is better than food.

$$t_{critical} = 1.90$$

Soln: $d = B - A = 3, 2, 1, 1, 3, 4, 2, 0$

$$\bar{d} = \frac{16}{8} = 2$$

$$SD = \sqrt{\frac{1}{7} [(3-2)^2 + (2-2)^2 + (1-2)^2 + (1-2)^2 + (3-2)^2 + (4-2)^2 + (2-2)^2 + (0-2)^2]}$$

$$= \sqrt{\frac{12}{7}} = 0.99$$

$$t = \frac{2 \times \sqrt{8}}{\sqrt{\frac{12}{7}}} = \frac{2 \times \sqrt{8} \times \sqrt{7}}{\sqrt{12}} = 4.32$$

$$t_{cal} > t_{critical}$$

⇒ H_0 is rejected

⇒ B is better than A

Q Below are the given gain in weights of dogs fed on two diets A & B.

A: 25, 32, 30, 34, 24, 14, 32, 24, 30, 31, 35, 25

B: 44, 34, 22, 10, 47, 31, 40, 30, 32, 35, 18, 21, 35, 29, 22

test if the 2 diets differ significantly as regard their effect on increase in weight, $t_{critical} = \pm 2.06$

Soln: $\bar{x}_1 = \frac{336}{12} = 28$, $\bar{x}_2 = \frac{450}{15} = 30$

$$s_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1} = \frac{\sum (x - 28)^2}{12 - 1} = \frac{380}{11} = 34.55$$

$$s_2^2 = \frac{\sum (x - \bar{x})^2}{n_2 - 1} = \frac{\sum (x - 30)^2}{15 - 1} = \frac{1410}{14} = 100.71$$

$$t_{cal} = \frac{28 - 30}{\sqrt{\frac{34.55}{12} + \frac{100.71}{15}}} = \frac{-2}{3.10} = -0.65$$

$$-2.06 < -0.65 < +2.06$$

⇒ H_0 is accepted

⇒ diff in A & B is not significantly at this level